Common Style & Syntax Issues

“Religious wars” I don’t care about

- Spacing / indentation / alignment
- Capitalization

```
SELECT …
FROM …
INNER JOIN …
WHERE … AND …
```

```
SELECT      …
FROM        …
INNER JOIN  …
WHERE       …
            AND …
```

```
select  …
from    …
   inner join …
where   …
   and  …
```

```
select …
  from …
       inner join …
where  …
   and …
```

Just be consistent.

Avoid implicit syntax defaults

Good

```
SELECT p.prod_name
FROM Manufacturer AS m
INNER JOIN Product AS p ON p.prod_manufacturer = m.man_name
WHERE m.man_state = 'TX'
```

Bad

```
SELECT p.prod_name
FROM Manufacturer AS m
JOIN Product p ON p.prod_manufacturer = m.man_name
WHERE m.man_state = 'TX'
```

Avoid implicit joins

Good

```
SELECT p.prod_name
FROM Manufacturer AS m
INNER JOIN Product AS p ON p.prod_manufacturer = m.man_name
WHERE m.man_state = 'TX'
```

Bad

```
SELECT p.prod_name
FROM Product AS p, Manufacturer AS m
WHERE p.prod_manufacturer = m.man_name AND
      m.man_state = 'TX'
```

Avoid turning join conditions into filter conditions

Good

```
SELECT p.prod_name
FROM Manufacturer AS m
INNER JOIN Product AS p ON p.prod_manufacturer = m.man_name
WHERE m.man_state = 'TX'
```

Bad

```
SELECT p.prod_name
FROM Product AS p
CROSS JOIN Manufacturer AS m
WHERE p.prod_manufacturer = m.man_name AND
      m.man_state = 'TX'
```



Explicitly list attributes

Good

```
SELECT p.prod_name, p.prod_price, p.prod_manufacturer
FROM Product AS p
```

Bad

```
SELECT *
FROM Product AS p
```

Always use table renaming

Good

```
SELECT p.prod_name
FROM Manufacturer AS m
INNER JOIN Product AS p ON p.prod_manufacturer = m.man_name
```

Bad

```
SELECT prod_name
FROM Manufacturer
INNER JOIN Product ON prod_manufacturer = man_name
```

```
SELECT Product.prod_name
FROM Manufacturer
INNER JOIN Product
    ON Product.prod_manufacturer = Manufacturer.man_name
```

Most important naming issues

- Be consistent.
- Make names meaningful.
- Make FK relationships clear.
- Avoid reusing names.
- Avoid overly long names.
- Avoid special names that need to be quoted.
  - Spaces and special characters
  - Keywords

Table design

- PK attributes placed first.

- Follow design rules of next module.

Remember unusual SQL syntax

- Strings use single quotes, not double 'Houston'
- Equality test uses single =, not double x = 5

Beware of web & text editors

Might get special non-ASCII characters.

'Houston' ‘Houston’

Two good style guides

http://www.sqlstyle.guide/

https://www.xaprb.com/blog/2006/04/26/sql-coding-standards/